



Enhancing RAGs for Rheumatology Triage: Strategies for Optimized Knowledge Retrieval

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Abstract. Managing the triage of chronic patients with rheumatic diseases is a significant challenge for general practitioners. Limited consultation time and a high volume of patients make it difficult to accurately identify those requiring specialist care, often leading to unnecessary referrals to specialists. Large language models can be integrated with retrieval-augmented generation to develop a generative AI system that assists general practitioners in identifying criteria for referring patients to rheumatology specialists, delivering fast and clinically reliable responses grounded in validated medical sources. To improve diagnostic efficiency while reducing the risk of errors, designing a curated knowledge base representation and a highly accurate retriever is indispensable. This study explored various retrieval strategies, including hybrid similarity computation methods, query augmentation, and noise-filtering techniques leveraging keywords and document structure. For initial tests, the knowledge base was populated with articles selected and validated by specialists from ICS Maugeri Hospital in Pavia, Italy. Preliminary results indicate that a hybrid lexical-semantic retrieval approach is the most effective for accessing the rheumatology knowledge base. Additionally, straightforward filtering strategies based on keywords and document structure can significantly reduce query times while minimizing noise.

Keywords: NLP · Clinical Decision Support · Rheumatic Diseases · RAG

1 Introduction

Triage for chronic patients with rheumatic diseases poses a significant challenge for general practitioners (GPs), as time constraints and high patient volumes complicate the identification of individuals requiring specialist evaluation. The need for technological solutions to support decision-making and improve triage accuracy is therefore evident. AI systems based on large language models (LLMs) coupled with retrieval-augmented generation (RAG) offer a valuable tool for assisting GPs in determining appropriate referrals to rheumatology specialists [1]. In knowledge-intensive tasks like GP triage,

the conversational nature of LLMs can speed and ease interactions, while the RAG framework guarantees that responses are anchored in validated medical sources [2]. However, the effectiveness of such a system depends heavily on the design of the knowledge base and the retrieval mechanism. The retriever’s role is crucial, as it extracts relevant knowledge from external sources based on the user query, directly determining the trustworthiness of the LLM answer. In this study, strategies for advanced retrieval design for rheumatology triage have been explored, partnering with rheumatology specialists from ICS Maugeri Hospital in Pavia, Italy, within the context of the DHEAL-COM (Digital Health Solutions in Community Medicine) project.

2 Methods

2.1 Knowledge Base and Evaluation Set

The rheumatology knowledge base was populated with nine peer-reviewed articles on rheumatoid arthritis classification criteria, focusing on clinical signs and symptoms. These articles were selected, curated and validated by rheumatology specialists. The documents’ structure has been preserved into logical hierarchical sections (i.e., chapters, subchapters, paragraphs, etc.) to facilitate retrieval. For evaluation, medical professionals formulated 75 questions related to referral criteria for rheumatology, linking each question to the relevant guideline excerpts, as shown by the example in Table 1.

Table 1. Example of questions and related knowledge base excerpts.

Question	Knowledge Base Excerpts
A patient with RA has more than three swollen joints and morning stiffness lasting 40 min. Is it appropriate to refer them to a rheumatologist?	Observations indicate that it is advisable to promptly refer a patient to a rheumatologist when rheumatoid arthritis (RA) is suspected. This recommendation is supported if at least one of the following criteria is met: more than three swollen joints, involvement of the metatarsophalangeal/metacarpophalangeal joints, and morning stiffness lasting more than 30 min

2.2 Syntactic-Semantic Documents Chunking and Representation

A syntactic-semantic chunking strategy was adopted. Initially, paragraph lengths were analyzed. If a paragraph exceeded 512 tokens, it was split into sentences. This was done to comply with BERT-based sentence embedders, limited to 512 input tokens. Consecutive sentences were then compared using cosine similarity on sentence embeddings, and sentences showing a similarity exceeding a specific threshold were grouped into a single chunk. The similarity threshold was dynamic, adapting to paragraph length based

on two distinct thresholds: a low threshold, active when the forming chunk is below 200 tokens, which encourages sentence aggregation, and a high threshold, triggered when the chunk size exceeds 200 tokens, aggregating following sentences only if strongly related to the chunk. Each new sentence is compared to the embedding of the entire forming chunk, allowing for more contextually coherent chunk formation. Each chunk was represented using both lexical (keywords) and semantic (embedding) features. For lexical representation, we considered two keyword extraction methods: YAKE! [3] and BM25. Ultimately, we chose YAKE! due to its corpus-agnostic nature. For semantic representation, we considered both biomedical [4] and a general-purpose sentence embedder for Italian, selecting the latter after preliminary comparisons¹.

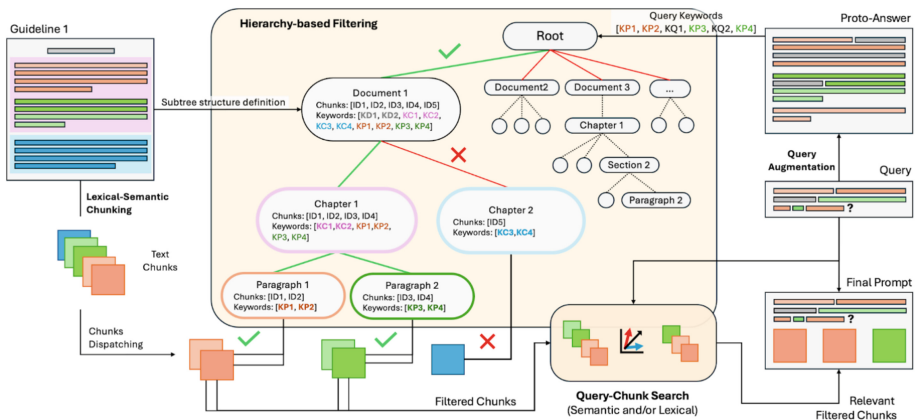


Fig. 1. Overview of the retrieving strategy implemented. On the left, a guideline from the knowledge base is split into five chunks using lexical-semantic chunking. The guideline’s structure is converted into a tree and appended to the root node. Each chunk is associated at every of the tree level with its corresponding node. On the right side, the user query undergoes query augmentation. Keywords are extracted and compared with those of nodes at a higher hierarchical level. If match is low, that branch is not inspected further (e.g., Chapter 2 and the corresponding blue chunk). The remaining filtered chunks are then compared with the query using lexical-semantic search, the most computationally expensive step. Finally, the most relevant filtered chunks are appended to the final prompt, which is passed to the LLM for answer generation. Color figure online

2.3 Query-Chunk Match: Hybrid Search and Query Augmentation

To enhance the retrieval of relevant documents given a query, we employed two techniques. The first combines a lexical score, based on keyword matching, with a semantic score derived from dense vector representations. These scores are fused using a convex parameter $\alpha \in [0,1]$. When α is zero, retrieval relies solely on keyword coverage, while when $\alpha = 1$, chunks and query are compared by pure cosine similarity between embeddings. The second technique builds on the idea that directly comparing a query

¹ huggingface.co/nickprock/sentence-bert-base-italian-uncased.

to a document is inherently mismatched [5]. Query and documents, even when strictly related, have very different formulations, styles, and lengths. To address this difference, the query is expanded through LLM zero-shot generation, producing a proto-answer for comparison. The proto-answer, generated by a general-purpose LLM (GPT-4o in our case) without additional instructions, is not grounded in a knowledge base but is structurally closer to the documents.

2.4 Filtering Using Document Hierarchy to Reduce Noise and Speed up Search

When searching large document collections, hierarchy-based filtering improves both efficiency and relevance. By structuring documents into logical levels, searches can be narrowed before performing expensive retrieval operations, reducing computation time and minimizes noise. We employed hierarchical indexing by leveraging the natural structure of document paragraphs. Each chunk is assigned to the corresponding paragraph node. At every level of the hierarchy, nodes inherit keywords from their children and are assigned five keywords derived from the combined text of all their descendants. During retrieval, query keywords are first compared against the highest-level nodes. Only nodes surpassing a predefined keyword coverage threshold (i.e., filtering intensity) are further inspected, ensuring that deeper searches occur only when relevant.

3 Results and Discussion

The hybrid retrieval system, illustrated in Fig. 1, was evaluated by running a five-fold cross-validation over the evaluation set, varying the alpha parameter and the filtering intensity between 0 and 1, for both original queries and augmented queries. The retrieving performance was assessed using the hit ratio metric (hits@k), which measures the proportion of correctly retrieved document chunks within the top-k results. In our setup, k is set to 5. The results of the 5-fold cross-validation show a hit ratio of 0.80 ± 0.08 . The best results are achieved by augmenting the query (5/5 folds), without applying filtering (0.00 ± 0.00), and with an alpha of 0.91 ± 0.07 , suggesting that a hybrid approach with a stronger emphasis on semantic search is beneficial and highlighting the added value of combining multiple representations, as already emerged in previous studies. Query augmentation significantly improved semantic retrieval, but had a marginal impact on lexical retrieval, as illustrated in Fig. 2. Filtering threshold analysis showed that moderate filtering, around 0.1–0.15, reduced retrieval time without affecting accuracy too much, making it advantageous for large-scale implementations.

4 Conclusion

This study highlights the effectiveness of improved retrieval strategies in RAG systems for supporting general practitioners in rheumatology triage. By optimizing a combination of lexical and semantic retrieval methods with query augmentation, the system provides reliable medical information while maintaining computational efficiency. On our test set, the retriever achieved a hit rate of up to 87%. Despite these improvements, certain

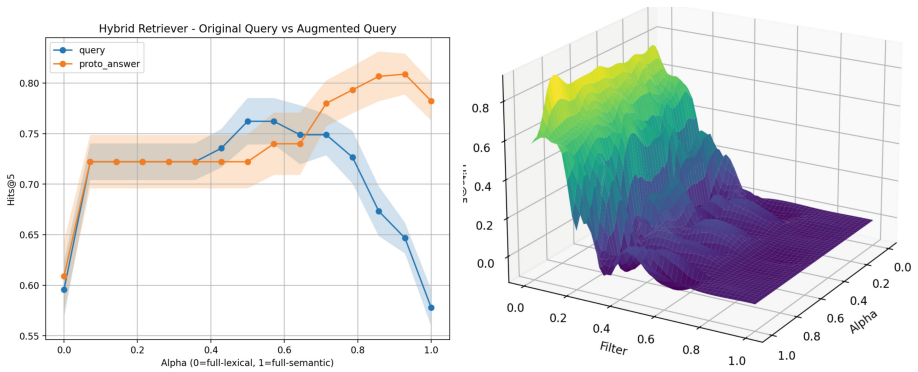


Fig. 2. Hybrid retriever results. On the left, retriever performance with and without query augmentation when varying alpha from full lexical (keywords only) to full-semantic (embeddings only). On the right, the full grid search when varying alpha and filter intensity.

limitations remain. The system does not yet incorporate multimodal data commonly found in clinical guidelines, such as medical images, tables, and flowcharts. Specialists note that flowcharts are highly informative and frequently used in rheumatology triage decisions. Additionally, integrating patient history alongside guidelines could enhance the system’s ability to provide personalized recommendations. Addressing these gaps will be a key focus of future work. Looking ahead, the most promising refinement lies in integrating knowledge graphs, which could leverage structured concepts and relationships to navigate across document chunks more effectively.

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References

1. Knitz, J., Gupta, L., Hügle, T.: Rheumatology in the digital health era: status quo and quo vadis? *Nat. Rev. Rheumatol.* 1–13 (2024)
2. Liu, S., McCoy, A.B., Wright, A.: Improving large language model applications in biomedicine with retrieval-augmented generation: a systematic review, meta-analysis, and clinical development guidelines. *J. Am. Med. Inform. Assoc.* ocaf008 (2025)
3. Campos, R., Mangaravite, V., Pasquali, A., Jorge, A., Nunes, C., Jatowt, A.: YAKE! keyword extraction from single documents using multiple local features. *Inf. Sci.* **509**, 257–289 (2020)

4. Buonocore, T.M., Crema, C., Redolfi, A., Bellazzi, R., Parimbelli, E.: Localizing in-domain adaptation of transformer-based biomedical language models. *J. Biomed. Inform.* **144**, 104431 (2023). <https://doi.org/10.1016/j.jbi.2023.104431>
5. Gao, L., Ma, X., Lin, J., Callan, J.: Precise zero-shot dense retrieval without relevance labels. *ArXiv Prepr* [arXiv:2212.10496](https://arxiv.org/abs/2212.10496) (2022)